Homework 2: Divisibility and Proofs

The CypherStack Team

Instructions

The purpose of this homework assignment is to review the concepts covered in Lecture 2 and introduce ideas that we did not have the time to cover. In the case that new material appears here that was not covered in class, please make note of any new notation or results, as they will be used in later lectures.

Please attempt each problem. Even if you are not successful, it is important to at least write down your thoughts on the problem and note where you are getting stuck, as well as any points of confusion.

New Material

Definition 1 (Modular Notation)

It is a bit of a pain to write the set $\{0, 1, 2, 3, ..., n-1\}$ every time we talk about values modulo n. To make life easier, we will introduce the notation:

$$\mathbb{Z}_n = \{0, 1, 2, ..., n-1\}$$

We refer to this as the group of integers modulo n.

Definition 2 (Modular Inverses)

Let $a \in \mathbb{Z}_n$. The inverse of a, denoted a^{-1}, is the element of $\mathbb{Z}_n$ such that:

$$a \cdot a^{-1} \equiv 1 \mod n$$

It is important to note that $a^{-1} \neq \frac{1}{a}$, since fractions don't exist in $\mathbb{Z}_n$. In general, note that $a^{-1} \bmod n$ is asking you to find x such that:

$$a \cdot x \equiv 1 \bmod n$$

Problems

Problem 1 (Modular Exponents)

(a) Compute $5^6 \bmod 7$

(b) Compute $2^36 \bmod 63$

(c) Find x such that $6^x \equiv 10 \bmod 11$.

(d) Was the prior question more or less difficult than the first two? Assuming the numbers in all three examples are extremely large, which of the problems feels like it would be the most difficult and why?

Problem 2 (Inverses)

Find each inverse or explain why none exists.

(a) $7^{-1} \bmod 12$

(b) $5^{-1} \bmod 13$

(c) $11^{-1} \bmod 14$

(d) Does 5 have an inverse mod 10?

Problem 3 (Divisibility)

Prove or disprove. For disproofs a single counterexample suffices; for proofs, full sentences.

(a) If $a \mid b$ and $a \mid (b + c)$, then $a \mid c$.

(b) If $a \mid c$ and $b \mid c$, then $ab \mid c$.

(c) If $a \mid bc$, then $a \mid b$ or $a \mid c$.

(d) If $\gcd(a, b) = 1$ and $a \mid c$ and $b \mid c$, then $ab \mid c$.

Problem 4 (Proof Methods)

For each, say which proof technique you would reach for *and why.*

(a) There is no largest prime.

(b) If n is an integer with n^3 odd, then n is odd.

(c) For every integer n, exactly one of n or $n + 1$ is even.

(d) $\gcd(a, b) = \gcd(a, b + ka)$ for every integer k.

If you are feeling particularly motivated, try to actually reason through the proofs. Try to use the arguments we went over in the lecture to write down a proof.